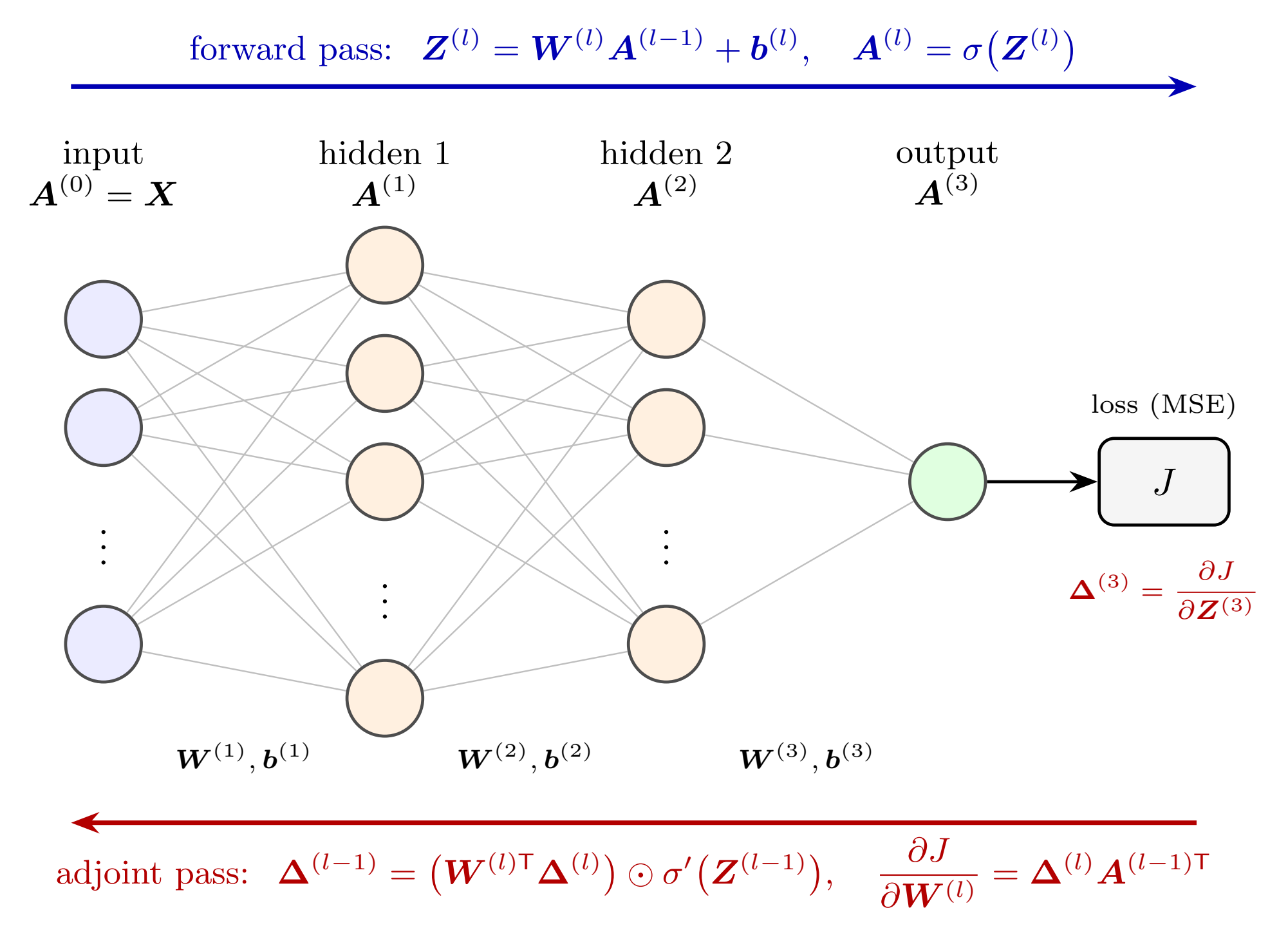

forward pass: $\boldsymbol{Z}^{(l)} = \boldsymbol{W}^{(l)}\boldsymbol{A}^{(l-1)} + \boldsymbol{b}^{(l)}, \quad \boldsymbol{A}^{(l)} = \sigma\big(\boldsymbol{Z}^{(l)}\big)$
input
$\boldsymbol{A}^{(0)} = \boldsymbol{X}$
hidden 1
$\boldsymbol{A}^{(1)}$
hidden 2
$\boldsymbol{A}^{(2)}$
output
$\boldsymbol{A}^{(3)}$
loss (MSE)
J
$\boldsymbol{\Delta}^{(3)} = \dfrac{\partial J}{\partial \boldsymbol{Z}^{(3)}}$
$\boldsymbol{W}^{(1)}, \boldsymbol{b}^{(1)}$
$\boldsymbol{W}^{(2)}, \boldsymbol{b}^{(2)}$
$\boldsymbol{W}^{(3)}, \boldsymbol{b}^{(3)}$
adjoint pass: $\boldsymbol{\Delta}^{(l-1)} = \big(\boldsymbol{W}^{(l)\mathsf{T}}\boldsymbol{\Delta}^{(l)}\big) \odot \sigma'\big(\boldsymbol{Z}^{(l-1)}\big), \quad \dfrac{\partial J}{\partial \boldsymbol{W}^{(l)}} = \boldsymbol{\Delta}^{(l)}\boldsymbol{A}^{(l-1)\mathsf{T}}$